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DriveNow ERP
Car Rental Management Platform

Business Documentation

DriveNow Car Rental Services

ITSAR2 - EndTerm Project

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1. Executive Summary

DriveNow ERP is a custom Enterprise Resource Planning platform built for **DriveNow Car Rental Services**, a single-branch car rental business in Bacolod City, Philippines. It digitizes and connects the company's core operations - fleet, reservations, customers, billing, maintenance and analytics - behind a unified, role-secured system, and adds a public self-service portal for customers. The solution is delivered as a microservices platform, satisfying the capstone brief's requirement for an independently-deployable, database-isolated, containerized architecture with a single secured entry point.

This document presents the **business** view: the company and its goals, the problems the ERP solves, the stakeholders and their permissions, the requirements, and the end-to-end operational processes the system implements. Each business process is cross-referenced to its technical implementation (see the companion Technical Documentation).

Aspect	Summary
Business	DriveNow Car Rental Services (single branch)
Location	Brgy. Alijis, Bacolod City, Negros Occidental, Philippines
Currency / Timezone	Philippine Peso (PHP) / Asia/Manila
System type	Microservices ERP + public customer portal
Modules	Fleet, Rentals, Customers (CRM), Billing, Maintenance, Analytics, Client Portal (+ Auth Hub)
User roles	Administrator, Staff, Maintenance Staff, Customer
Status	Built and running under Docker Compose; pending public deployment + demo

2. Business Overview

Car rental is a logistics-and-service business: a finite fleet of vehicles must be matched to customer demand over time, kept road-worthy, billed accurately, and analyzed for profitability. DriveNow's operations span six interlocking functions - keeping the fleet ready (Fleet, Maintenance), turning demand into revenue (Rentals, Billing), nurturing the customer base (CRM, Client Portal), and steering the business with data (Analytics).

Historically these functions are handled with disconnected tools (spreadsheets, paper logs, ad-hoc records). The ERP replaces them with a single source of truth per function, connected by well-defined interfaces, so that - for example - a vehicle's availability, its active rental, its invoice and its maintenance history are all consistent and visible to the right staff.

3. Company Profile

Field	Detail
Company name	DriveNow Car Rental Services
Industry	Car rental / ground transportation services
Size	Single branch; small operational team (administration, front-desk, maintenance)
Location	Brgy. Alijis, Bacolod City, Negros Occidental, Philippines
Contact	hello@drivenow.test | +63 34 000 0000 | https://drivenow.test
Mission	To provide reliable, affordable vehicle rentals for Bacolod City and nearby areas, backed by efficient, transparent operations.

> These details are the platform's configured business identity (the Operations Hub *settings*), which brand every service at runtime. An administrator can change the business name, logo, currency and timezone centrally and all services follow.

4. Business Objectives

1. **Centralize operations** into one connected platform with a single sign-on for staff.
2. **Maximize fleet utilization** by making vehicle availability and status visible across rentals and maintenance.
3. **Accelerate the rental cycle** from reservation to return with a clear, enforced workflow.
4. **Ensure billing accuracy** with itemized invoices, multiple payment methods, penalties and refunds.
5. **Strengthen customer relationships** through profiles, loyalty tiers, feedback and a self-service portal.
6. **Enable data-driven decisions** via consolidated analytics and exportable reports.
7. **Enforce accountability and security** through role-based access and an activity audit trail.

5. Problem Statement

Operating a rental business with disconnected tools creates recurring operational pain:

Pain point	Consequence
Vehicle availability not shared across functions	Double-bookings, idle vehicles, manual cross-checking
Manual, inconsistent reservation handling	Errors, slow turnaround, no clear status of each booking
Billing done separately from rentals	Revenue leakage, disputed charges, missed penalties
Maintenance tracked on paper	Overdue servicing, unsafe vehicles, no cost visibility
Customer data scattered	No loyalty insight, repeat-customer friction, no blacklist control
No consolidated reporting	Decisions made without utilization or revenue insight
Shared logins / no access control	No accountability; sensitive actions not restricted

The ERP is designed to directly resolve each of these pain points, as mapped in Section 24 (Benefits).

6. Proposed Solution

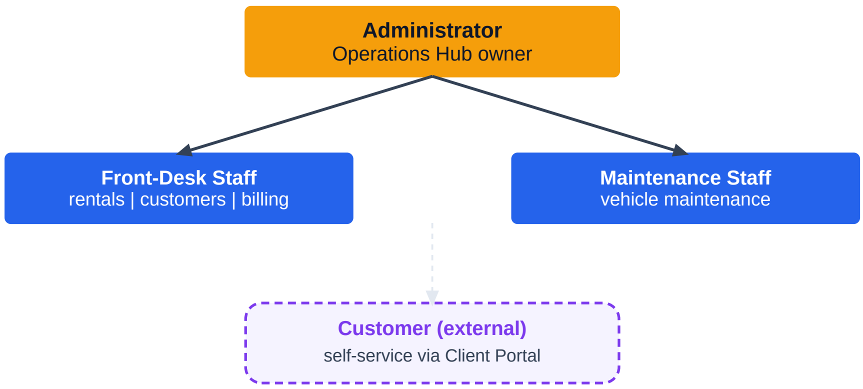
DriveNow ERP addresses these problems with **seven functional modules plus an authentication hub**, each owning its data and exposing a secured interface, unified by single sign-on and a consistent design. Staff work in role-appropriate admin applications; customers use a public portal.

Module	Solves	Key capabilities
Operations Hub (Auth)	Access control, branding	Login/SSO, roles, system settings, user management, activity log
Fleet	Availability visibility	Vehicle inventory, categories, status, daily rates, photos
Rentals	Reservation handling	Booking with conflict check, approval, release, return, cancel, extend
Customers (CRM)	Scattered customer data	Profiles, loyalty tiers/points, feedback, blacklist, history
Billing	Billing accuracy	Invoices (penalty/discount), payments (cash/GCash/card), refunds
Maintenance	Paper maintenance logs	Scheduled/repair/inspection/damage records, workflow, cost
Analytics	No reporting	KPI dashboards, date-range reports, CSV export
Client Portal	Customer friction	Public catalogue, self-service booking, profile management

7. Stakeholder Analysis

Stakeholder	Interest in the system	Primary modules
Administrator / Owner	Overall control, configuration, oversight, reporting	All modules + settings/users
Front-Desk Staff	Fast, accurate day-to-day operations	Rentals, Customers, Billing, Fleet
Maintenance Staff	Keep vehicles serviced and safe	Maintenance (+ fleet status)
Customers	Easy browsing and self-service booking	Client Portal
The business	Profitability, utilization, accountability	Analytics, audit log

Organizational Structure - roles that interact with the ERP



Internal staff use the admin services (:8001-8007); customers use the public portal (:8008).

Figure 7-1. Organizational structure and how each role interacts with the platform.

8. User Roles and Responsibilities

The system defines four roles (the `UserRole` enumeration: **admin, staff, maintenance, customer**). Authorization is enforced in software: all staff areas require authentication, and every data-changing action is additionally gated to the appropriate role(s).

8.1 Responsibilities

- **Administrator -** full access to all modules; manages global system settings and staff user accounts; the only role that can change configuration and create/deactivate users.
- **Front-Desk Staff -** processes reservations, manages customers, issues invoices and records payments, and maintains the vehicle inventory.
- **Maintenance Staff -** creates and progresses vehicle maintenance records.
- **Customer -** uses the public portal only: browses vehicles, makes and tracks their own bookings, and manages their own profile.

8.2 Roles and Permissions Matrix

The matrix below is derived directly from the implemented route authorization (the `role` middleware gates). "Yes" = create/update/delete permitted; "Read" = view only; "Own" = limited to the user's own records; "-" = no access.

Capability	Admin	Staff	Maintenance	Customer
System settings & branding	Yes	-	-	-
Staff user management	Yes	-	-	-
Fleet / vehicles	Yes	Yes	Read	-
Rentals / reservations	Yes	Yes	Read	-
Customers (CRM)	Yes	Yes	Read	-
Billing / invoices & payments	Yes	Yes	Read	-
Maintenance records	Yes	Read	Yes	-
Analytics & reports	Read	Read	Read	-
Browse vehicles (portal)	-	-	-	Yes
Create / track bookings (portal)	-	-	-	Own
Manage own profile (portal)	-	-	-	Own

Verified: a customer-role identity attempting any back-office mutation is rejected with **HTTP 403**; read/dashboard pages remain available to authenticated staff. Maintenance staff are scoped to maintenance changes; front-desk staff to operational modules; only administrators may change settings or users.

9. Business Requirements

The system must support the complete operational cycle of a single-branch rental business while honouring the capstone's architectural mandates.

- BR-1 Maintain an accurate, shared record of every vehicle and its current status.
- BR-2 Capture reservations and move them through a controlled lifecycle to completion.
- BR-3 Prevent double-booking of a vehicle for overlapping dates.
- BR-4 Produce accurate invoices and record customer payments by multiple methods.
- BR-5 Track vehicle maintenance and keep unsafe vehicles out of circulation.
- BR-6 Maintain customer profiles, loyalty and feedback, and allow blacklisting.
- BR-7 Provide consolidated operational and financial reporting.
- BR-8 Let customers self-serve (browse and book) online.
- BR-9 Restrict every sensitive action to authorized roles and keep an audit trail.

10. Functional Requirements

ID	Requirement	Module
FR-1	Add, edit, categorize and set the status/daily rate of vehicles	Fleet
FR-2	Create reservations with a vehicle picker drawn from live availability	Rentals
FR-3	Reject reservations that overlap an existing booking for the same vehicle	Rentals
FR-4	Approve, release (pick-up), return, extend and cancel reservations	Rentals
FR-5	Generate invoices with subtotal, penalty and discount; track balance	Billing
FR-6	Record payments (cash / GCash / card) and process refunds	Billing
FR-7	Maintain customer profiles, loyalty tier/points, feedback and blacklist	CRM
FR-8	Schedule and progress maintenance records by type and severity	Maintenance
FR-9	Display KPI dashboards and a date-range report with CSV export	Analytics
FR-10	Allow customers to register, browse, book and manage their bookings	Client Portal
FR-11	Authenticate users and enforce role-based authorization (SSO)	Auth Hub
FR-12	Provide global, centrally-managed branding/currency/timezone	Auth Hub

11. Non-Functional Requirements

Category	Requirement (as implemented)
Architecture	Microservices; one isolated database per service; single API gateway entry point
Security	Token-based SSO (JWT); role-based authorization; service-token-protected internal APIs; CSRF; password hashing; optional 2FA/passkeys
Reliability	Graceful degradation on peer failure; health endpoints; container restart policy
Usability	Consistent responsive UI, dark/light mode, toasts, confirmation dialogs, accessible modals
Maintainability	Uniform per-service structure; config-driven cross-service links; versioned APIs
Portability	Fully containerized; one-command Docker Compose bring-up
Localization	Philippine Peso currency and Asia/Manila timezone, centrally configurable
Auditability	Uniform activity logging across services; on-screen activity feed in the Hub

12. Business Process Workflows

This section documents the platform's principal end-to-end processes (the brief requires at least five). Each is implemented as an explicit, role-gated workflow in the corresponding service; the states shown are the actual domain enumerations.

12.1 Vehicle / Fleet management

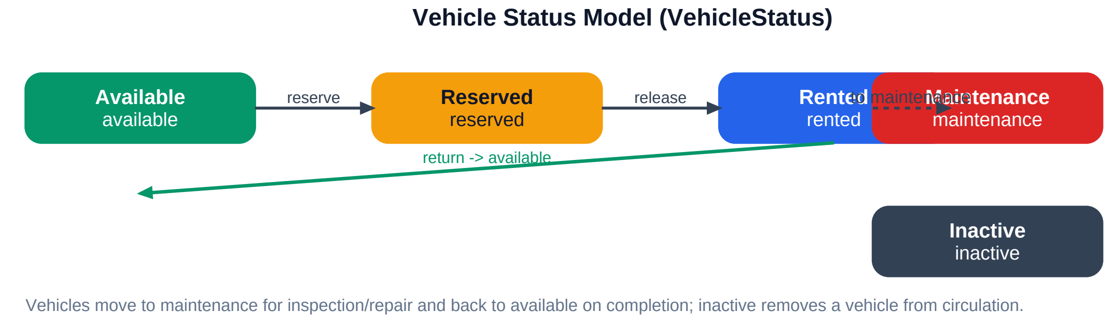


Figure 12-1. Vehicle status model and transitions.

12.2 Rental reservation lifecycle

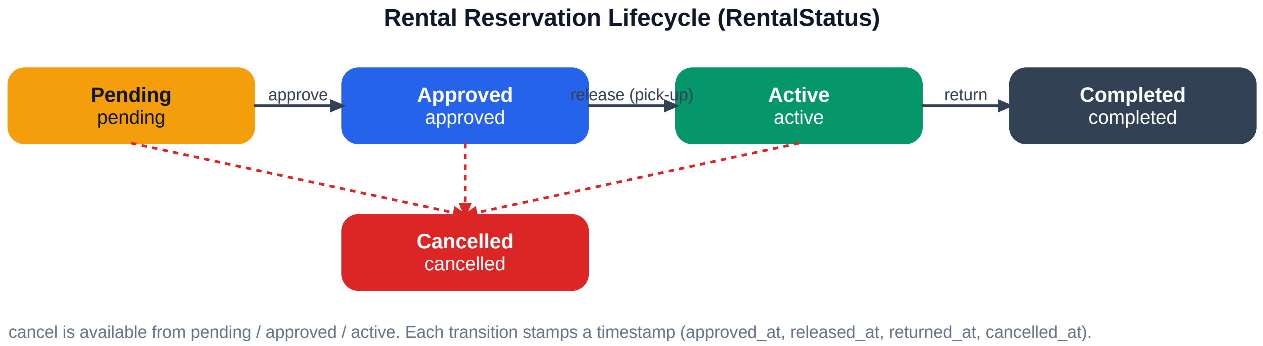


Figure 12-2. Reservation lifecycle from pending to completed (or cancelled).

12.3 Invoicing & payment

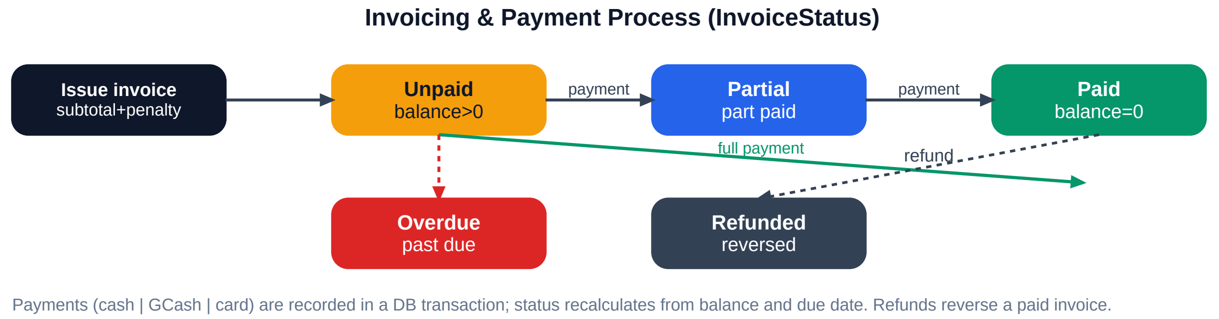


Figure 12-3. Invoice status driven by payments, due date and refunds.

12.4 Maintenance

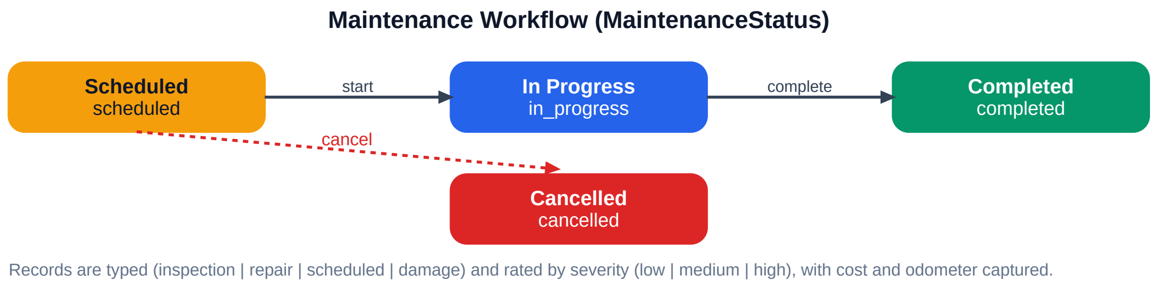


Figure 12-4. Maintenance record workflow.

12.5 Customer management

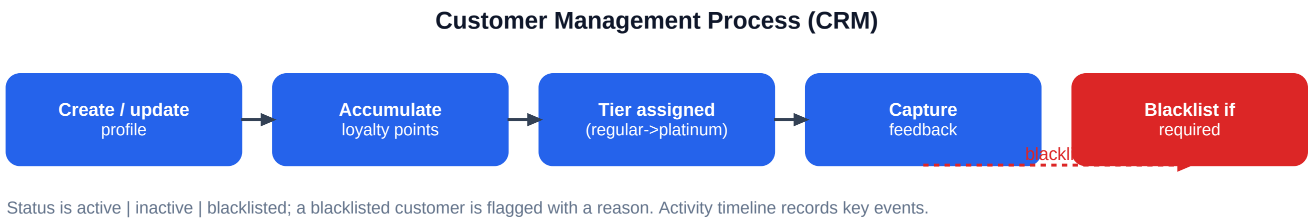


Figure 12-5. Customer lifecycle in CRM.

12.6 Customer self-service booking

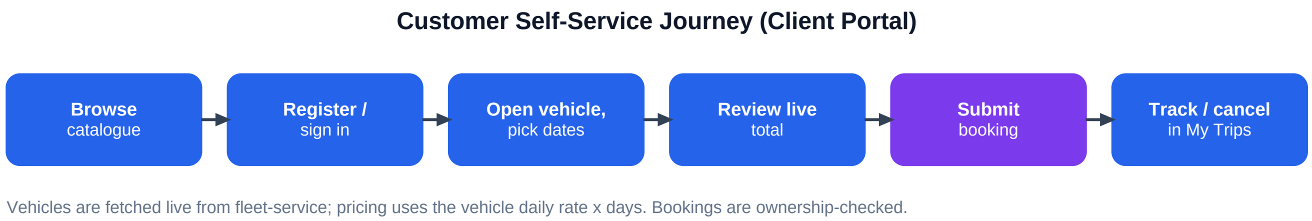


Figure 12-6. Public customer journey through the Client Portal.

13. Use Case Diagrams

The actors are the four roles; the system boundary encloses the capabilities each actor can invoke. Administrators have the broadest reach; staff inherit operational use cases; maintenance is scoped to maintenance; customers act only through the portal.

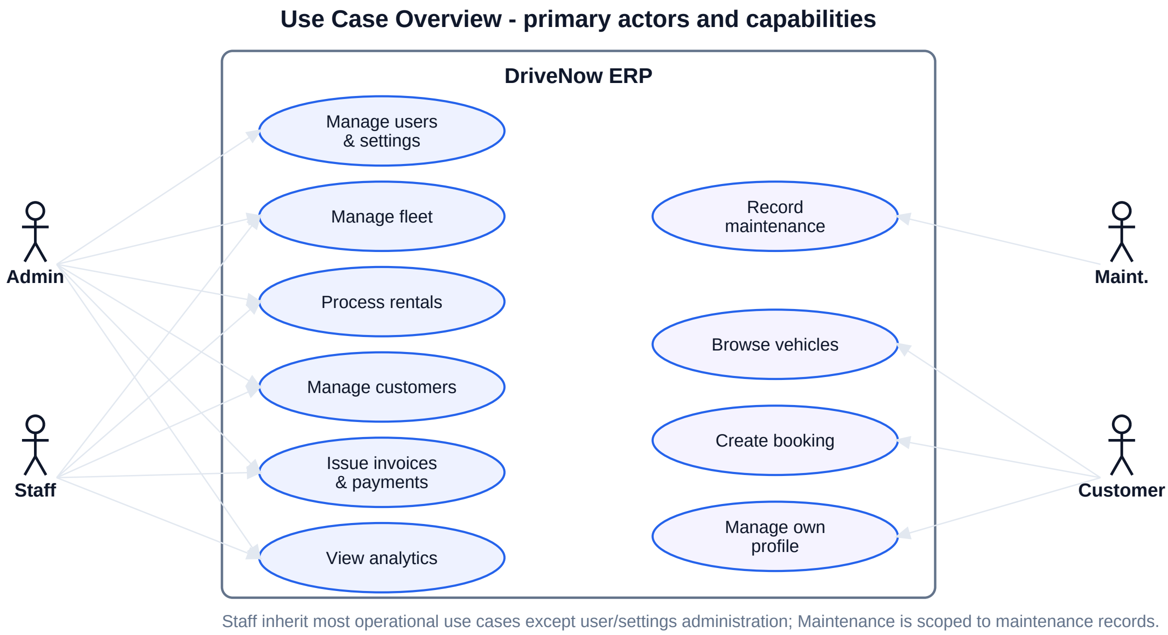


Figure 13-1. Primary use cases by actor.

14. User Journey Maps

14.1 Customer (self-service booking)

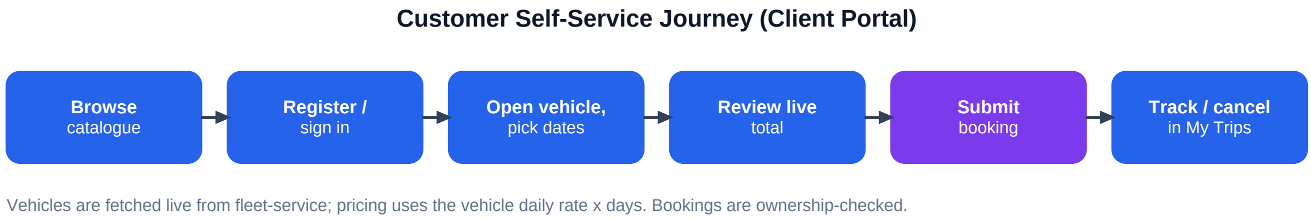


Figure 14-1. Customer journey: browse, authenticate, select, review, book, track.

14.2 Staff (process a walk-in rental)

1. Open Rentals and start a new reservation.
2. Pick an available vehicle (list fetched live from Fleet) and enter the customer and dates.
3. The system validates there is no date conflict and computes the total.
4. Approve the reservation, then release it at pick-up.
5. On return, mark it returned; Billing issues the invoice and records payment.

14.3 Maintenance staff (service a vehicle)

1. Open Maintenance and create a record for the vehicle (type and severity).
2. Start the work (status: in progress); the vehicle is kept out of circulation.
3. Complete the record with cost and odometer; the vehicle returns to available.

15. Operational Procedures

- **Daily start**: staff sign in once at the Operations Hub; SSO grants access to permitted modules.
- **Reservation intake**: create the booking, confirm availability, approve.
- **Pick-up**: release the reservation; the vehicle becomes rented.
- **Return**: mark returned; raise the invoice; record payment; apply penalties if applicable.
- **Maintenance**: schedule or log repairs; vehicles under maintenance are unavailable for booking.
- **End of period**: review analytics dashboards and export reports.
- **Administration**: manage staff accounts and global settings as needed (admin only).

16. Rental Process Lifecycle

The reservation is the heart of the business. Its lifecycle is an explicit state machine; each transition is a deliberate, role-gated action that records a timestamp.

State	Meaning	Entry action	Next
Pending	Booking created, awaiting approval	create reservation	Approved / Cancelled
Approved	Confirmed, awaiting pick-up	approve	Active / Cancelled
Active	Vehicle released to customer	release (pick-up)	Completed / Cancelled
Completed	Vehicle returned	return	(terminal)
Cancelled	Booking voided	cancel	(terminal)

Before a reservation is accepted, the system checks for date overlaps against existing bookings of the same vehicle and computes the total from the daily rate and number of days. Reservations can also be extended. Downstream, Billing references the rental and Analytics counts it toward revenue and utilization.

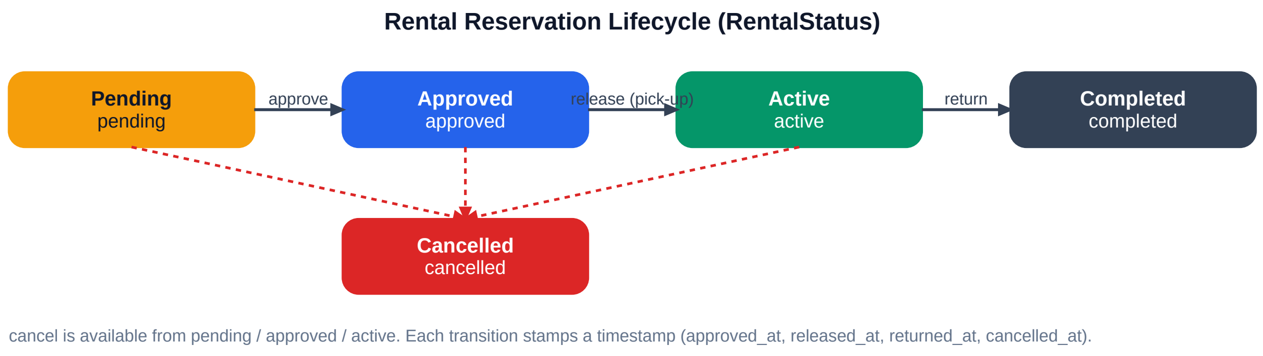


Figure 16-1. Reservation state machine (RentalStatus).

17. Customer Management Process

CRM maintains a single profile per customer with contact details, status, loyalty standing, feedback and an activity timeline. Loyalty **tier** (regular, silver, gold, platinum) is derived from accumulated points; **status** is active, inactive or blacklisted, with a recorded reason when blacklisted. A customer's rental history is read from the Rentals service.

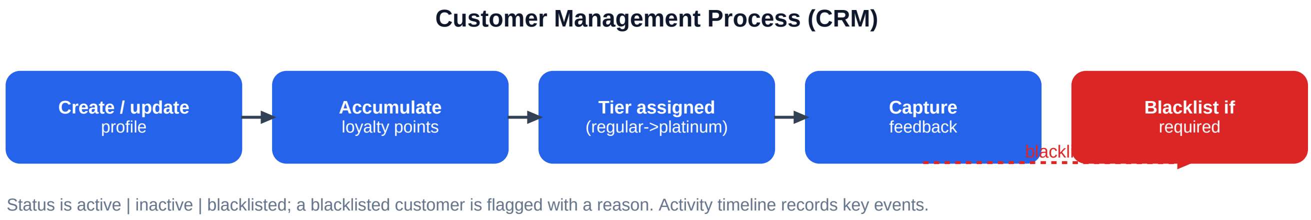


Figure 17-1. Customer lifecycle and key CRM actions.

18. Vehicle Management Process

Fleet is the authoritative inventory. Each vehicle carries make, model, year, plate, category (sedan, hatchback, SUV, van, pickup), branch, daily rate, seats, transmission, fuel type, mileage and an optional photo. **Status** (available, reserved, rented, maintenance, inactive) governs whether a vehicle can be booked and is the signal other modules rely on.

Vehicle Status Model (VehicleStatus)

Vehicles move to maintenance for inspection/repair and back to available on completion; inactive removes a vehicle from circulation.

Figure 18-1. Vehicle status transitions.

19. Reporting & Analytics Overview

The Analytics module gives management a consolidated view without manual collation. It aggregates indicators by reading the other services and renders dashboards plus a date-range report that can be exported to CSV.

- **Revenue** trends over time.
- **Fleet utilization** and top-performing vehicles.
- **Rental trends** (volume, status mix).
- **Payment mix** (by method) and **customer retention** indicators.
- **Maintenance** activity and cost overview.

> Analytics is read-only. When a source service is temporarily unreachable it clearly flags the dashboard as showing representative (degraded) figures rather than presenting incomplete data as final.

20. Policies and Business Rules

#	Business rule (enforced in software)
BR-A	A vehicle cannot be reserved for dates that overlap an existing booking of that vehicle.
BR-B	Rental totals are computed from the vehicle's daily rate multiplied by the rental days.
BR-C	Reservations progress only through valid transitions (pending -> approved -> active -> completed); cancellation is allowed before completion.
BR-D	Invoice status is recalculated from the outstanding balance and due date (unpaid, partial, paid, overdue, refunded).
BR-E	Payments are accepted by cash, GCash or card and recorded atomically.
BR-F	Vehicles under maintenance or marked inactive are not available for booking.
BR-G	Blacklisted customers are flagged with a reason for staff attention.
BR-H	Only administrators may change global settings or manage staff accounts.
BR-I	Deactivated staff accounts cannot sign in; the final administrator cannot be removed or demoted (anti-lockout).
BR-J	Every sensitive (data-changing) action is restricted to the authorized role(s).

21. Risk Assessment

Risk	Likelihood	Impact	Mitigation (in system / planned)
Double-booking a vehicle	Low	High	Date-overlap conflict check before acceptance (implemented)
Unauthorized data changes	Low	High	Role-based authorization on all mutations (implemented)
Revenue leakage / billing error	Low	High	Invoices tied to rentals; atomic payments; status recalculation (implemented)
Unsafe / overdue vehicle	Medium	High	Maintenance workflow + status keeps vehicles out of service (implemented)
Service outage affecting others	Low	Medium	Graceful degradation + health checks (implemented)
Stale access after deactivation	Low	Medium	Login blocked immediately; token expires <=2h; denylist planned
Data loss of uploads on rebuild	Medium	Medium	Use a shared volume / object storage (planned)
No public URL by deadline	Medium	Medium	Stack is deploy-ready; single VPS bring-up remaining

22. Benefits and Expected Outcomes

Pain point (Section 5)	Expected outcome with DriveNow ERP
Availability not shared	Real-time vehicle status visible to rentals and maintenance; fewer conflicts
Manual reservations	Guided, validated booking workflow; faster, error-resistant turnaround
Separate billing	Invoices tied to rentals; accurate charges, penalties and payment tracking
Paper maintenance	Digital records, scheduling and cost visibility; safer fleet
Scattered customers	Unified profiles, loyalty and feedback; informed service and retention
No reporting	Consolidated dashboards and exportable reports for decisions
No access control	Role-based security and an audit trail; accountability

23. System Scope and Limitations

23.1 In scope (implemented)

- All eight services (Auth Hub + six admin modules + Client Portal), running under Docker Compose.
- Role-based SSO, internal API security, and the workflows documented above.
- Centralized branding/currency/timezone and an activity audit feed.

23.2 Out of scope / Not Implemented

- **Public production URL** - the stack is deploy-ready but not yet hosted publicly (planned).
- **Online payment gateway integration** - payment *methods* are recorded (cash/GCash/card) but no external payment processor is integrated; recording is manual.
- **Automated email/SMS notifications** - Not Implemented.
- **Native mobile app -** the portal is responsive web; no native app.
- **Kubernetes orchestration** (a brief bonus) **-** Not Implemented.
- **Automated test suite / CI-CD** - Not Implemented (see Technical Documentation §22-23).

> Where a capability is partially present it is described precisely above to avoid overstating scope. The documentation reflects the current state of the system, not assumed future functionality.

24. Implementation Strategy

The platform was built incrementally, module by module, on a shared service template, with each service verified (type-check, build, migrate/seed, route listing, formatting) before moving on. The remaining work to reach full capstone completion is primarily deployment and documentation.

Phase	Scope	Status
1	Auth Hub foundation (roles, settings, SSO)	Done
2-3	Fleet and Rentals modules	Done
4	CRM, Billing, Maintenance, Analytics	Done
5	Client Portal	Done
6	Cross-service SSO hardening (JWT), API versioning, service tokens	Done
7	Containerization + Docker Compose + gateway	Done
8	Security/role hardening, deploy-readiness, documentation	In progress
9	Public deployment + demo video + presentation	Planned

25. Training Requirements

The interface is intentionally consistent across modules, so role-focused orientation is brief:

Audience	Focus	Indicative effort
Administrators	Settings, user management, reading the audit feed, all modules	Half-day
Front-desk staff	Reservation lifecycle, customers, invoicing & payments	Half-day
Maintenance staff	Creating and progressing maintenance records	1-2 hours
Customers	Self-service portal (no training; guided UI)	None

> Formal training materials (manuals, videos) are **Not Available** yet; this document and the planned demo video serve as the initial reference.

26. Support & Maintenance Process

- **Health monitoring**: each service exposes a health endpoint; the Operations Hub shows live online/offline status of every module.
- **Incident handling**: per-service container logs and a uniform activity log support diagnosis; services degrade gracefully so one fault does not halt the platform.
- **Configuration changes**: branding, currency and timezone are changed centrally without a redeploy.
- **Releases**: code changes are rolled out by rebuilding the affected service image (Docker).
- **Data**: each service's database can be backed up independently; demo data can be re-seeded on a fresh environment.

> A formal SLA, ticketing process and on-call rotation are organizational decisions and are **Not Defined** in the current project scope.

27. Future Business Enhancements

1. **Public deployment** with a real domain so customers can use the portal online.
2. **Online payments**: integrate a payment processor (e.g., GCash/card gateway) so the recorded payment methods become real transactions.
3. **Notifications**: automated booking/return/maintenance reminders by email or SMS.
4. **Promotions & dynamic pricing**: seasonal rates and loyalty-based discounts.
5. **Multi-branch support**: extend the (already-present) branch field into full multi-location operations.
6. **Customer mobile app** and richer self-service (modify/extend bookings online).
7. **Deeper analytics**: forecasting and profitability per vehicle/segment.

> These are forward-looking and explicitly **not yet implemented**. The remainder of this document describes only functionality present in the current system.